

BUBT Intra University Programming Contest–BIUPC Summer 2026 (Junior) Mock

<https://toph.co/c/bubt-intra-university-biupc-summer-2026-junior-mock>



Schedule

The contest will run for **5h0m0s**.

Authors

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Rules

This contest is formatted as per the official rules of ICPC Regional Programming Contests.

You can use C++17 GCC 13.2, C++20 Clang 16.0, C++20 GCC 13.2, C++23 GCC 13.2, C17 GCC 13.2, C23 GCC 13.2, Java 1.8, Kotlin 1.9, Kotlin 2.0, PyPy 7.3 (3.10), and Python 3.12 in this contest.

Be fair, be honest. Plagiarism will result in disqualification. Judges' decisions will be final.

Notes

There are 12 challenges in this contest.

Please make sure this booklet contains all of the pages.

If you find any discrepancies between the printed copy and the problem statements in Toph Arena, please rely on the latter.

A. Eid Mubarak

Eid is one of the most joyous occasions for Muslims around the world. Every year, Muslims celebrate two major Eids according to the Islamic (Hijri) calendar, each carrying a unique message of gratitude, faith, and compassion.

The first is Eid-ul-Fitr (ঈদ উল ফিতর), celebrated on 1 Shawwal. It marks the end of the blessed month of Ramadan---a month devoted to fasting, worship, patience, and self-discipline. Eid-ul-Fitr is a day of immense gratitude to Allah, filled with joy, generosity, and spending cherished time with family and friends.

The second is Eid-ul-Adha (ঈদ উল আজহা), celebrated on 10 Zil Hajj. It commemorates the profound test of faith faced by Prophet Ibrahim (عليه السلام). After remaining childless for over 80 years, Ibrahim (عليه السلام) was finally blessed with a son, Ismail (عليه السلام). Years later, when Ismail (عليه السلام) had grown into a youth capable of working alongside his father, Ibrahim (عليه السلام) faced an immense test. The command from Allah to sacrifice Ismail (عليه السلام) came to Ibrahim (عليه السلام) in a dream. As a Prophet, Ibrahim (عليه السلام) knew that the dreams of the Messengers are true revelations from Allah, meaning there was no choice but to completely comply. When Ibrahim (عليه السلام) shared this command, Ismail (عليه السلام) demonstrated ultimate devotion and replied, "O my father, do whatever you are ordered. You will find me among the patient."

Both father and son willingly submitted to the command of Allah. However, just as Ibrahim (عليه السلام) was about to perform the sacrifice, Allah accepted Ibrahim's absolute devotion. Out of boundless mercy, Allah replaced Ismail (عليه السلام) with a sheep. It was a large, horned, white male sheep which was sent from Jannah.

To honor this unparalleled act of submission, Muslims worldwide celebrate Eid-ul-Adha by sacrificing an animal. The meat is beautifully divided into three equal parts: one for the family, one for relatives, and one to be given to the poor and needy, spreading compassion throughout the community.

Your junior friend has learned about these two beautiful occasions but still gets confused about which date belongs to which Eid. Help your friend by writing a program that identifies the correct Eid!

Input

The input consists of a single line containing a string representing a date in the Islamic (Hijri) calendar.

It is guaranteed that the input is either "1 Shawwal" or "10 Zil Hajj".

Output

Print “Eid-ul-Fitr” if the input is “1 Shawwal”; Print “Eid-ul-Adha” if the input is “10 Zil Hajj”.

Example

Input	Output
1 Shawwal	Eid-ul-Fitr

B. Copycat

Read an integer variable and print it.

Input

The input will contain an integer A ($-2000000000 < A < 2000000000$).

Output

Print the integer.

Example

Input	Output
2	2

C. Add Them Up

Read two integer variables, calculate their sum, and print it.

Input

The input will contain two integers A and B ($-20000000 < A, B < 20000000$).

Output

Print the sum of the two integers.

Example

Input	Output
4 5	9

D. Formatted Numbers

Read an integer variable and print it in which the digits are separated into groups of three by commas.

Input

The input will contain an integer A ($0 \leq A < 2000000000$).

Output

Print the formatted number.

Example

Input	Output
1171123	1,171,123

E. Divisors

Read an integer variable and print all of its divisors (including 1 and the number itself).

Input

The input will contain an integer A ($0 < A < 100$).

Output

Print the divisors in increasing order, one per line.

Example

Input	Output
6	1 2 3 6

F. Fibonacci Numbers

Given an integer N , print the N -th Fibonacci number.

A Fibonacci series is a series of numbers in which each number is the sum of the two numbers preceding it. For this problem, you can assume that the first two numbers in the series are 1 and 1.

Below you can see the first few numbers of the series:

1, 1, 2, 3, 5, 8, ...

Input

The input will contain an integer N ($0 < N < 50$).

Output

Print the N -th Fibonacci number.

Example

Input	Output
3	2

G. B-Number

Bruce Lee has recently become the sensei (Master) of the very institute which taught him everything. He has found out a way to be unique to his students. After every class of him starts, he says "I am back." Also, before leaving the class, he says, "I will be back." Dollar, a naughty but brilliant student of Master Lee, has decided to make a notebook of his class which will only consist of these two lines. So, after a semester, his notebook look something like this.

```
I will be back.  
I am back.  
I will be back.  
I am back.  
I will be back.  
.....  
.....  
I am back.
```

You saw Dollar's notebook after the semester ended and wondered how many classes of Lee did Dollar attend in the whole semester. When you asked Dollar, he said you need not count the times he wrote these two sentences. He count the number of times the letter "b" appears in his notebook and wrote it in front of the notebook. He named it the B-number and said you can guess the number of classes attended by having a look at it. Can you really guess how many classes of Master Lee did he attend in that semester? Dollar told you three important things: (i) Master Lee did not say "I am back." in his first class (ii) he did not say "I will be back" in his last class. (iii) He attended at least two classes in a semester.

Input

The first line of input contains an integer T ($1 \leq T \leq 20000$). Each of next T lines contain an integer B ($1 \leq B \leq 2^{31}$) which denotes a B-number. Each line is separated with a newline.

Output

For each B (B-number) in input, print in separate lines how many classes Dollar attended to write this number. Each line will have a newline.

Examples

Input	Output
1 6	3

Input	Output
For the first sample, Dollar's notebook looked like this: I will be back. I am back. I will be back. I am back.	

Input	Output
2 9 102	4 35

H. GCD Plus LCM

In this problem you will be given two integers a, b . You have to answer if $GCD(a, b) + LCM(a, b) = a + b$ is true or not.

Input

Input begins with an integer T ($0 < T \leq 10^6$) for the number of cases to follow. Next T lines each contain two integers a and b ($0 < a, b \leq 10^{18}$).

Output

For each case print **true** or **false** without quotes according to the problem statement.

Example

Input	Output
3	true
2 10	false
2 3	false
4 6	

I. Clock Math

Given a time (hours as H and minutes as M), determine the smaller angle between the two hands of a clock showing the time and print it.



Input

The input will contain two integers: H ($0 \leq H < 12$) and M ($0 \leq M < 60$).

Output

Print the angle in degrees (positive and accurate to 10^{-4}).

Example

Input	Output
10 15	142.50000000

J. Big Factorials

Given an integer N , print the trailing 4 digits of $N!$ (N factorial).

$$N! = N \times (N - 1) \times (N - 2) \times \dots \times 1$$

Here are some examples of $N!$ and their last 4 digits.

N	N Factorial ($N!$)	Last 4 Digits
3	6	6
7	5040	5040
11	39916800	6800
15	1307674368000	8000

Input

The input will contain an integer N ($0 < N < 1000$).

Output

Print the last 4 digits of $N!$. In case $N!$ has less than 4 digits, do not print any leading zeros.

Examples

Input	Output
4	24

Input	Output
10	8800

K. Full Pyramid

Given an integer N , print a full pyramid of asterisks.

A full pyramid of asterisks of size N has N lines of asterisks. The first line has 1 asterisk, the second line has 2 asterisks, the third line has 3 asterisks and so on. Each asterisk has a space between them. The asterisks in each line are centered to make it look like a pyramid.

Here is a full pyramid of asterisks of size 4:

```
  *
 * *
* * *
* * * *
```

Input

The input will contain an integer N ($1 \leq N \leq 100$).

Output

Print the full pyramid of asterisks of the given size.

Do not print any space after the last asterisk in each line.

Example

Input	Output
3	<pre> * * * * * *</pre>

L. Byang's Additions

Byang is learning how to add numbers. However, he gets confused whenever there is a carry.

To help Byang you need to write a program that will read two integers and determine if Byang will encounter any carry when adding these two numbers.

For example when adding 182 and 243 Byang will encounter a carry:

```
  182
+ 243
-----
   5 (Carry 0)
  ---
  25 (Carry 1; This will confuse Byang)
  ---
 425 (Carry 0)
```

On the other hand, when adding 123 and 321 Byang will not encounter any carry, and it will not confuse Byang:

```
  123
+ 321
-----
   4 (Carry 0)
  ---
  44 (Carry 0)
  ---
 444 (Carry 0)
```

Input

The input will contain two space-separated integers A and B ($0 \leq A, B < 1000000$).

Output

Print "Yes" if adding A and B will confuse Byang, otherwise "No".

Examples

Input	Output
182 243	Yes

Input	Output
123 321	No